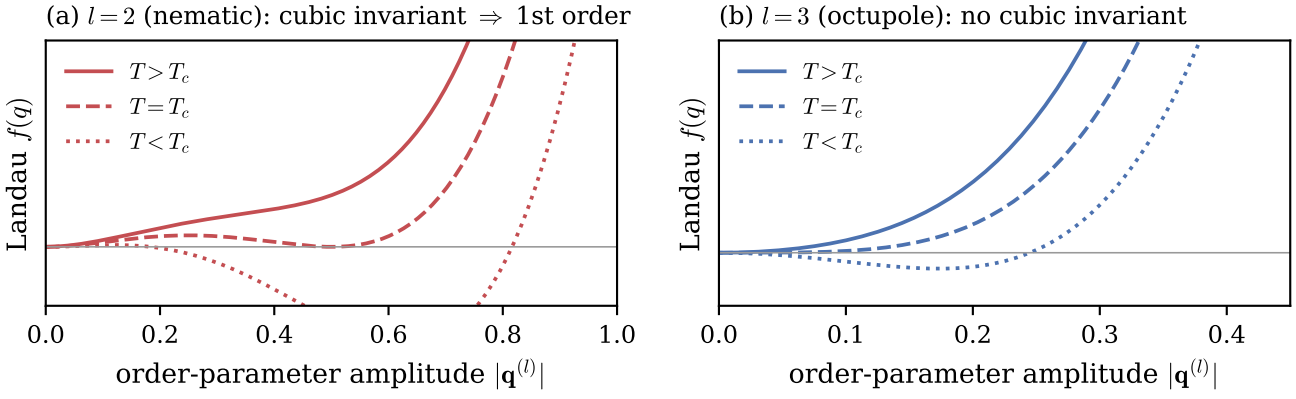
(a) $l = 2$ (nematic): cubic invariant $\Rightarrow$ 1st order
$T > T_c$
$T = T_c$
$T < T_c$
Landau $f(q)$
0.0
0.2
0.4
0.6
0.8
1.0
order-parameter amplitude $|\mathbf{q}^{(l)}|$
(b) $l = 3$ (octupole): no cubic invariant
$T > T_c$
$T = T_c$
$T < T_c$
Landau $f(q)$
0.0
0.1
0.2
0.3
0.4
order-parameter amplitude $|\mathbf{q}^{(l)}|$